



AUTONOMOUS FRAMEWORK FOR REAL-TIME DATA QUALITY, PRIVACY, AND SCHEMA DRIFT MANAGEMENT IN STREAMING ENVIRONMENTS

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GROUP MEMBERS



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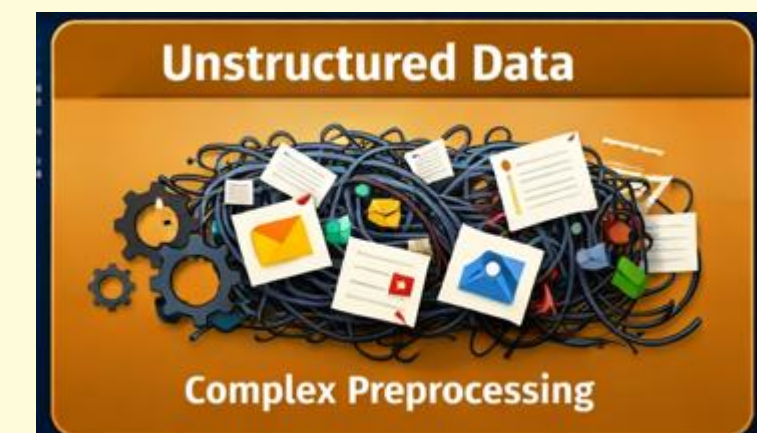
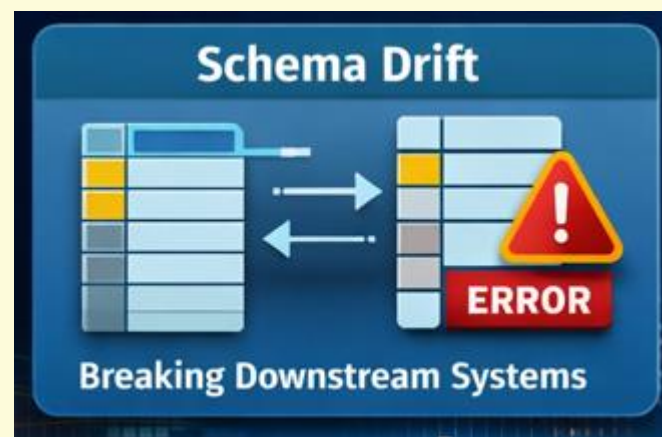
NANDUNI

Safeguarding Data Pipelines



Problem Statement

Modern organizations rely heavily on **streaming data pipelines** for analytics and decision making.



Impact

- Broken data pipelines
- Incorrect analytics results
- Privacy risks
- Increased manual intervention

Stakeholders



Data Engineers



Data Scientists



System Admins



Data Platform Architects

Research Gap

Existing systems mainly focus on **individual problems** and often handle them **after data is stored**.

They lack:

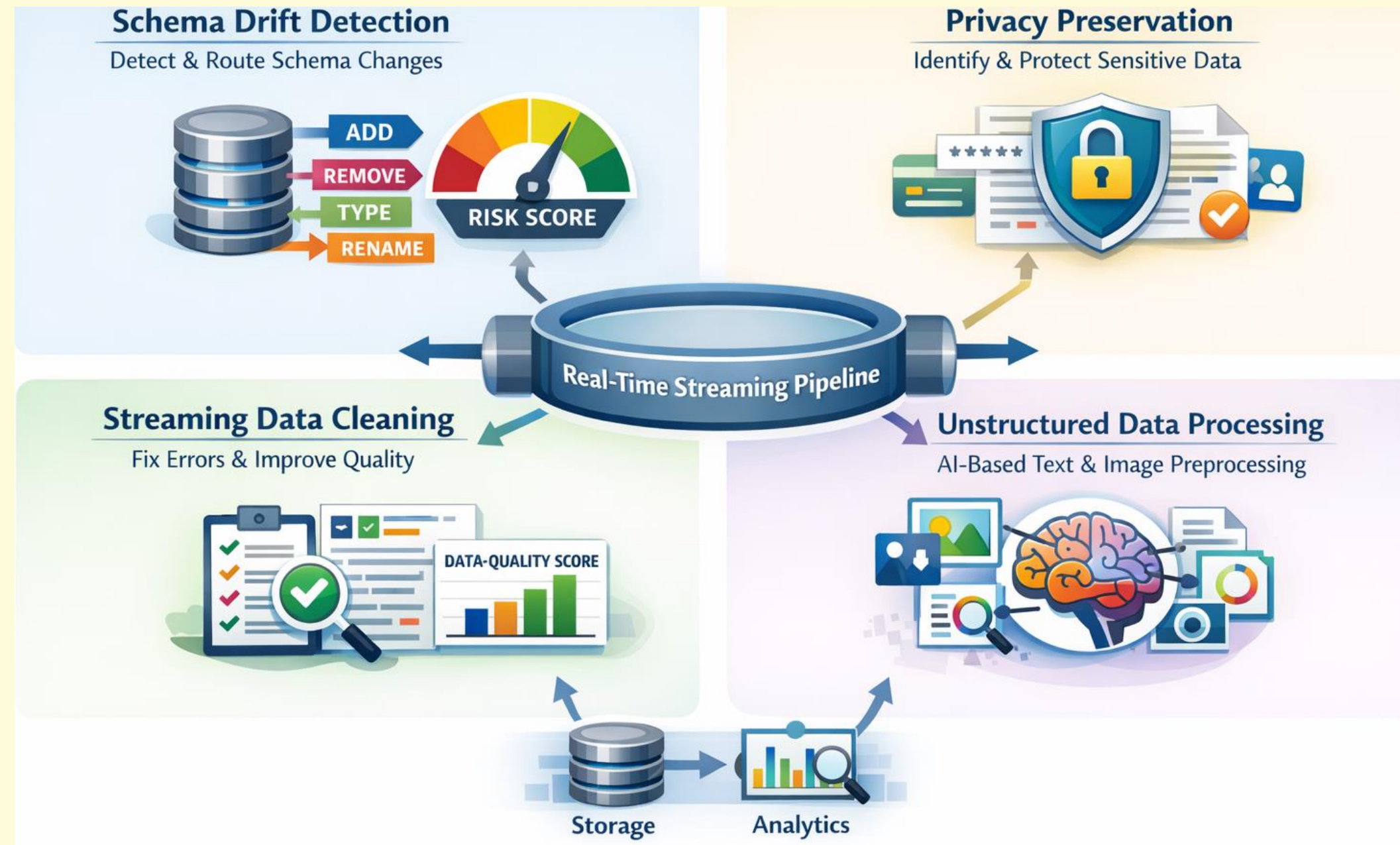
Integrated preprocessing for **multiple data issues**

Real-time streaming support

Context-aware privacy protection

Intelligent preprocessing for **unstructured data**

Main Objective



Solution Overview

A real-time intelligent preprocessing pipeline for streaming data

Key Features

- Real-time schema drift detection with risk scoring
- Automatic identification and protection of sensitive attributes
- Real-time data quality detection and cleaning
- Agentic AI preprocessing for text and images

Non-Functional Focus

- Performance (real-time streaming)
- Reliability (pipeline stability)
- Security (privacy protection)
- Scalability (Kafka-based architecture)

System Architecture

COMMERCIALIZATION & SUSTAINABILITY

Value Proposition

Prevent pipeline failures

Protect sensitive data automatically

Improve data quality before storage

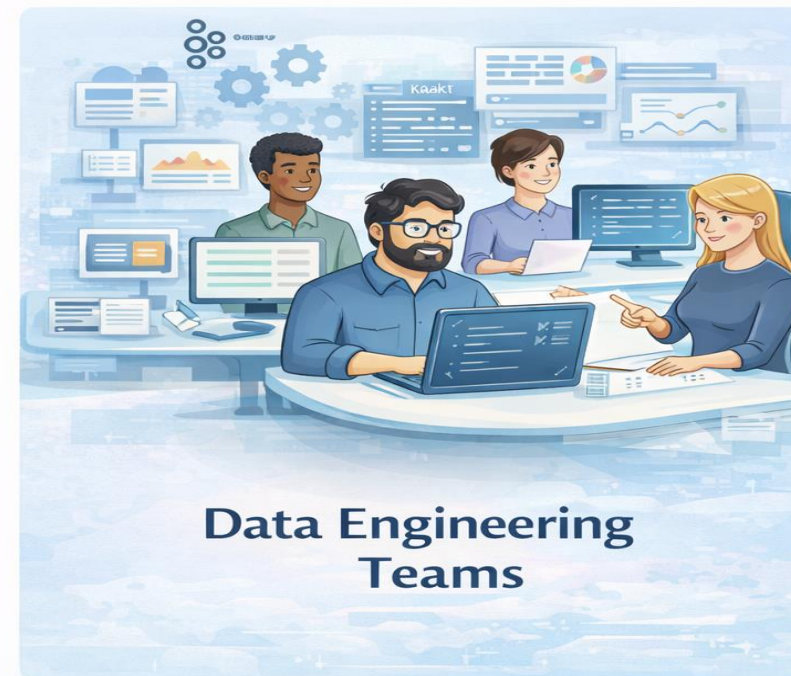
Enable reliable analytics

Feasibility

Built using scalable technologies

Target Users

platforms



Data Engineering Teams



Organizations Using Streaming Data Platforms

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THANK YOU